

OS LAB WEEK 9

Program to simulate Page Replacement Algorithms
FIFO,LRU,Optimal

```
#include <stdio.h>
```

```
int search(int frames[], int n, int page) {  
    for (int i = 0; i < n; i++)  
        if (frames[i] == page) return 1;  
    return 0;  
}
```

```
void print_frames(int frames[], int n) {  
    printf("[");  
    for (int i = 0; i < n; i++) {  
        if (frames[i] == -1) break;  
        if (i > 0) printf(" ");  
        printf("%d", frames[i]);  
    }  
    printf("]");  
}
```

```
void fifo(int pages[], int num_pages, int num_frames) {  
    int frames[num_frames];  
    for (int i = 0; i < num_frames; i++) frames[i] = -1;  
    int page_faults = 0, next = 0;  
  
    printf("\n--- FIFO Page Replacement ---\n");  
    for (int i = 0; i < num_pages; i++) {  
        printf("Page %d -> ", pages[i]);  
        if (!search(frames, num_frames, pages[i])) {  
            frames[next] = pages[i];  
            next = (next + 1) % num_frames;  
            page_faults++;  
        }  
        print_frames(frames, num_frames);  
        printf("\n");  
    }  
    printf("Total Page Faults (FIFO): %d\n", page_faults);  
}
```

```
void lru(int pages[], int num_pages, int num_frames) {  
    int frames[num_frames];  
    int last_used[num_frames];  
    for (int i = 0; i < num_frames; i++) { frames[i] = -1; last_used[i] = -1; }
```

```

int page_faults = 0;

printf("\n--- LRU Page Replacement ---\n");
for (int i = 0; i < num_pages; i++) {
    printf("Page %d -> ", pages[i]);
    int found = 0;
    for (int j = 0; j < num_frames; j++) {
        if (frames[j] == pages[i]) {
            last_used[j] = i;
            found = 1;
            break;
        }
    }
    if (!found) {
        int pos = -1;
        for (int j = 0; j < num_frames; j++) {
            if (frames[j] == -1) { pos = j; break; }
        }
        if (pos == -1) {
            int lru_time = last_used[0], lru_idx = 0;
            for (int j = 1; j < num_frames; j++) {
                if (last_used[j] < lru_time) { lru_time = last_used[j]; lru_idx = j; }
            }
            pos = lru_idx;
        }
        frames[pos] = pages[i];
        last_used[pos] = i;
        page_faults++;
    }
    print_frames(frames, num_frames);
    printf("\n");
}
printf("Total Page Faults (LRU): %d\n", page_faults);
}

```

```

void optimal(int pages[], int num_pages, int num_frames) {
    int frames[num_frames];
    for (int i = 0; i < num_frames; i++) frames[i] = -1;
    int page_faults = 0;

    printf("\n--- Optimal Page Replacement ---\n");
    for (int i = 0; i < num_pages; i++) {
        printf("Page %d -> ", pages[i]);
        if (!search(frames, num_frames, pages[i])) {
            int pos = -1;
            for (int j = 0; j < num_frames; j++) {
                if (frames[j] == -1) { pos = j; break; }
            }

```

```

    }
    if (pos == -1) {
        int farthest = -1, replace = -1;
        for (int j = 0; j < num_frames; j++) {
            int next_use = num_pages;
            for (int k = i + 1; k < num_pages; k++) {
                if (pages[k] == frames[j]) { next_use = k; break; }
            }
            if (next_use > farthest) { farthest = next_use; replace = j; }
        }
        pos = replace;
    }
    frames[pos] = pages[i];
    page_faults++;
}
print_frames(frames, num_frames);
printf("\n");
}
printf("Total Page Faults (Optimal): %d\n", page_faults);
}

```

```

int main() {
    int num_pages, num_frames;

    printf("Enter number of pages: ");
    scanf("%d", &num_pages);

    int pages[num_pages];
    printf("Enter page reference string:\n");
    for (int i = 0; i < num_pages; i++)
        scanf("%d", &pages[i]);

    printf("Enter number of frames: ");
    scanf("%d", &num_frames);

    fifo(pages, num_pages, num_frames);
    lru(pages, num_pages, num_frames);
    optimal(pages, num_pages, num_frames);

    return 0;
}

```

```
Enter number of pages: 12
Enter page reference string:
7 0 1 2 0 3 0 4 2 3 0 3
Enter number of frames: 3
```

```
--- FIFO Page Replacement ---
```

```
Page 7 -> [7]
Page 0 -> [7 0]
Page 1 -> [7 0 1]
Page 2 -> [2 0 1]
Page 0 -> [2 0 1]
Page 3 -> [2 3 1]
Page 0 -> [2 3 0]
Page 4 -> [4 3 0]
Page 2 -> [4 2 0]
Page 3 -> [4 2 3]
Page 0 -> [0 2 3]
Page 3 -> [0 2 3]
Total Page Faults (FIFO): 10
```

```
--- LRU Page Replacement ---
```

```
Page 7 -> [7]
Page 0 -> [7 0]
Page 1 -> [7 0 1]
Page 2 -> [2 0 1]
Page 0 -> [2 0 1]
Page 3 -> [2 0 3]
Page 0 -> [2 0 3]
Page 4 -> [4 0 3]
Page 2 -> [4 0 2]
Page 3 -> [4 3 2]
Page 0 -> [0 3 2]
Page 3 -> [0 3 2]
Total Page Faults (LRU): 9
```

--- Optimal Page Replacement ---

Page 7 -> [7]

Page 0 -> [7 0]

Page 1 -> [7 0 1]

Page 2 -> [2 0 1]

Page 0 -> [2 0 1]

Page 3 -> [2 0 3]

Page 0 -> [2 0 3]

Page 4 -> [2 4 3]

Page 2 -> [2 4 3]

Page 3 -> [2 4 3]

Page 0 -> [0 4 3]

Page 3 -> [0 4 3]

Total Page Faults (Optimal): 7